



Multispectral imaging through metasurface with quasi-bound states in the continuum

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Abstract: By controlling light fields in subwavelength scales, metasurfaces enable novel ways for miniaturization and integration of spectral imaging system. Metasurfaces supporting quasi bound states in the continuum (quasi-BICs) can control the quality factor and spectral response by changing structural parameters. In this work, we present an ultra-compact multispectral imaging device, whereby spectral modulation is achieved by meta-atoms arrays supporting quasi-BICs. The designed meta-atom array can serve as filters over a wide range of wavelengths, which enables the device capable of a large operating range and high-fidelity spectral reconstruction with a fine spectral resolution. The microspectrometers composed of BIC metasurfaces also can work as imaging pixels to achieve computational imaging spectroscopy through periodic arrangement, which successfully resolves images with spatial aliasing in different channels. This spectrometer device can meet the market demand for miniaturization for rapidly object recognition and appropriate spatial spectral resolution at low cost.

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1. Introduction

After light interacts with matter, spectra can reveal the chemical composition and physical property of substances, widely used in the fields of product safety [1,2], medical diagnosis [3,4], biological imaging [5,6] and agricultural inspection [7,8]. Traditional high-precision spectrometers are usually bulky, expensive due to the need for dispersion devices such as prisms [9] and gratings [10]. Especially when further applied in spectral imaging, it needs to be realized by line scanning or area scanning, which is not only time-consuming and inefficient, but also requires complex and precise mechanical moving parts, closely associated to its spatial resolution. With the increasing demand for device integration and size reduction, the research of compact spectrometer has become an important topic. In realistic applications, the tradeoffs between spectral resolution and bandwidth versus device cost and miniaturization should be considered in spectral imaging technology [11–13]. To miniaturize spectrometers, researchers have explored various advanced methods, one of which is to combine metasurfaces with computational optics [14–16].

Metasurfaces composed of artificial subwavelength units have provide a new method for wavefront control [17–19]. The designed meta structure can greatly enhance the interaction with light and allow people to arbitrarily control the amplitude, phase, polarization, frequency and other parameters of light [20–23]. Meanwhile, metasurfaces are ultra-compact and easy to integrate. Due to these advantages, metasurfaces have the potential to become a platform for the preparation of ultra-compact spectral imaging devices [24,25]. By controlling light field parameters in subwavelength scales, metasurfaces provide new opportunities for miniaturization

and integration of spectral imaging system. Compact micro-spectrometers integrated with metasurfaces have been designed and fabricated. Y. Zhou [26] studied the influences of structural parameters of meta-lens-based spectrometers on the effective spectral range and the spectral resolution and designed two types of meta-lens based spectrometers. M. Faraji-Dana [27] introduced the concept of folded metasurface optics by demonstrating a compact spectrometer and fabricated an ultra-compact line scanning miniature hyperspectral imager with a total product of only 8.5 mm. Y. Huang et al. [28,29] proposed a snapshot hyperspectral image sensor based on metasurfaces and applied it to anti-spoofing face recognition and rat dynamic brain spectrum acquired.

In order to improve the spectral detection resolution, ultrahigh quality factor (Q factor) is required. Recently, metasurfaces with bound states in the continuum (BICs) have also attracted much attention due to its infinite Q factor [30,31]. The concept of BIC was discovered in quantum mechanics for the first time [32], but was later expanded to optics. In practice, BICs are usually realized as quasi-BICs with high finite Q factor in the form of supercavity modes for highly spectrally selectivity enhancement [33]. Furthermore, the high Q resonances are spectrally clean and only occur at the selective wavelengths, which is particularly meaningful for removing unwanted background spectra without additional filters. Experimental realizations have demonstrated that the quasi-BIC with ultrahigh Q factor is very suitable for sensing and detection. For example, A. Tittl [34] developed an imaging-based nanophotonic method which featured a 2D pixelated dielectric metasurface with a range of ultrasharp resonances for detecting mid-infrared molecular fingerprints. F. Yesilkoy [6] combined dielectric metasurfaces to develop an ultrasensitive label-free analytical platform for biosensing. These research results demonstrate the spectral control capability of BIC metasurfaces, showing their potential on spectral imaging.

In this paper, we proposed a compact multispectral imager in the near infrared band based on quasi-BIC metasurfaces. The metasurface we designed contains nine types of meta-atom arrays with different periods. These meta-atom arrays support quasi-BICs with different eigenfrequencies due to period changes. Based on the frequency selective properties of BIC metasurfaces, each kind of meta-atom array functions as a band stop filter working at resonant frequency of supported quasi-BIC. The fabricated meta-atom array shows high reflection efficiency in the working frequency band and low crosstalk with neighboring channels. A superpixel composed of nine types of meta-atom arrays can work as a single shot compact spectrometer. By scaling a single super pixel to pixel arrays, the designed metasurface will work as a snapshot spectral imager. We have experimentally verified that spectral resolution of the proposed metasurface device can achieve 2.5 nm with fidelity above 92%. The spatially overlapping images were also separated from different wavelengths light projection. The proposed system based on non-local BIC metasurface shows excellent performance in snapshot spectral detection and imaging, indicating great application prospects in remote sensing, biosensing, etc.

2. Method

2.1. System design overview

The schematic of the proposed multispectral imaging system based on quasi-BIC metasurface is shown in Fig. 1. In our system, the metasurface consisting of quasi-BIC meta-atom arrays are placed a distance away from the sensor. The metasurface device can be seen as a coded mask and encodes the spectral information of the incoming light via an optical transmission matrix. The modulated light is received by the infrared camera and each filter corresponds to several pixels on the CMOS sensor. In this way, we are able to formulate an underdetermined equation and use algorithms to recover the spectral data cube or point mapping spectral curve of the image from a subset of sensor pixels. The spectral modulated matrix can be predetermined by monochromator calibration. Herein, we use a superpixel to illustrate the working principle of a single microspectrometer, as shown in Fig. 2(b). A superpixel is composed of meta-atom

filter arrays with different geometric parameters. These meta-atom arrays have quasi-BIC with different eigenfrequencies, resulting in different reflection spectra $R_i(\lambda)$, where λ is the free-space wavelength and i is the sequence number of atom arrays. Thus, the imaging process of the i th meta-atom array can be expressed as:

$$b_i = \int_{\lambda_1}^{\lambda_2} f(\lambda) R_i(\lambda) \eta(\lambda) d\lambda = \int_{\lambda_1}^{\lambda_2} f(\lambda) A_i(\lambda) d\lambda \quad (1)$$

where b_i is grayscale value of the i th meta-atom array recorded by CMOS, $f(\lambda)$ is the incident spectrum to be recovered, $\eta(\lambda)$ is the absorption quantum efficiency of the CMOS sensor, λ_1 and λ_2 are the lower and upper wavelength of the operational band range. However, the actual measurement is a discrete process and we set $R_i(\lambda)\eta(\lambda) = A_i(\lambda)$, where $A_i(\lambda)$ represents the i th row of the modulated matrix. Thus, the above integral Eq. (1) can be expressed as:

$$b_i = \sum_{j=1}^M f_j(\lambda) A_{ij}(\lambda) \quad (2)$$

where M is the number of wavelength bands, j is the j th wavelength band. In the proposed model, we demonstrated the modulation process of a spatial point spectrum signal. The incident spectral signal will be reconstructed from the grayscale image captured by CMOS sensor.

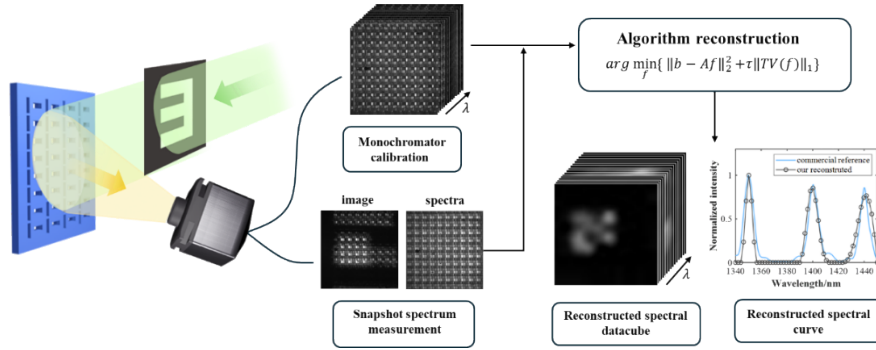


Fig. 1. Overview of the Metasurface Spectral imaging pipeline, which reconstructs a hyperspectral datacube from a snapshot spectrum measurement.

2.2. Design and characterization of devices

BIC is a special type of resonance state localized in the structure and coexist with a continuous spectrum of radiating waves [35]. The true BIC is a theoretical object which can only be realized in an ideal lossless structure with an infinite period region. When a structural symmetry perturbation is applied, the BIC transforms to quasi-BIC with limited Q factor, regulated by changing the degree of structural asymmetry perturbation [36,37]. In our design, we achieved the quasi-BIC states by a pair of orthogonal rectangular air holes. Figures 2(a) and (b) show schematics of the unperturbed and perturbed meta-atom. When the length of one edge of the square lattice changes, the period in real space doubles in one spatial direction, resulting in the folding of the first Brillouin band. The bound modes are brought into continuum and coupled due to the perturbation. It has been shown that the coupling strength is related to the magnitude of the perturbation. By these construction, a symmetry-protected BIC is obtained, whose Q factor is driven by the following formula for small values of δ :

$$Q = C/\delta^2 \quad (3)$$

where C is a constant number, depending on the period, geometric size, polarization and modes [37]. As shown in Fig. 2(c) we have already proved that C can be presented with fitting parameter

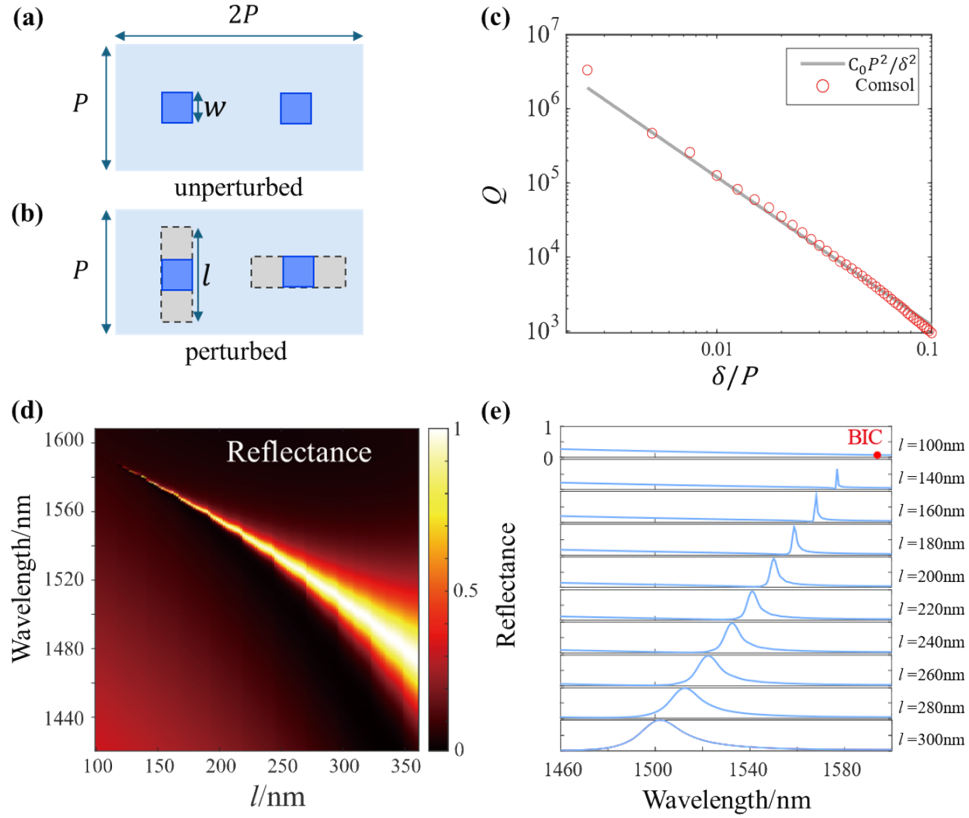


Fig. 2. The principles and characteristics quasi-BICs. Schematics of the unperturbed (a) and perturbed (b) meta-atom, where $P = 400$ nm and $w = 100$ nm are period and length of the unperturbed lattice respectively. (c) Dependence of the Q factor on the perturbation, $\delta = l - w$. (d) Eigenmode spectra and reflectance spectra with respect to resonance wavelength and length l . (e) The spectral characteristics of BIC with the varied perturbation factor.

C_0 and period P in this structure. Figures 2(d) and (e) reveal the evolution of a true BIC to a quasi-BIC by scaling the asymmetric parameter about l . The reflection narrow peaks gradually vanish with the growth of radiation loss caused by perturbation. The symmetry protected BIC with infinity Q factor exist when $l = w$, as shown by the red dot in Fig. 2(e). The BIC converts to quasi-BIC with finite Q factor when square holes become rectangular holes. Through these simulations of this model, it is proved that BIC exists in the designed structure.

Our device is composed of an array of all-dielectric metasurface device, allowing for multiplex spectral detection in a single measurement. Based on the above principle simulation, we chose a pair of air holes with a fixed perturbation, and adjusted the horizontal period p_x of structures uniformly to obtain filters with almost the same resonance linewidth and different resonance frequencies. Each air hole is a rectangle with a length l of 300 nm and a width w of 100 nm. By geometrically scaling p_x from 680 nm to 760 nm with a step of 10 nm, our metasurface can be spectrally tuned. The vertical period p_y is still fixed at 400 nm. Specifically, we use a 3 by 3 spectral filter array to build a microspectrometer [Fig. 3(b)], of which each unit is a meta-atom array mapped by tens of CMOS pixels in imaging. Every meta-atom array performs as a spectral filter corresponding to a resonance wavelength from 1340 nm to 1450 nm. One filter area has a size of $16 \mu\text{m} \times 16 \mu\text{m}$, so that the superpixel size is $50 \mu\text{m}$ with adjacent filters spaced at

a distance of $1\mu\text{m}$. Our metasurface was fabricated by depositing a 250 nm thick amorphous Si film ($h=250\text{ nm}$) on a glass substrate followed by electron beam lithography. Scanning electron microscopy (SEM) images of the meta-atoms and superpixels are shown in Fig. 3(e). In Fig. 3(b), we use different grayscale values to represent meta-atoms distribution density, which can intuitively reflect changes of the period p_x , and its numerical values are indicated on the right colorbar. When we normally incident a beam of x-polarized light, collective lattice resonances will be excited due to symmetry breaking of highly symmetric resonator geometries.

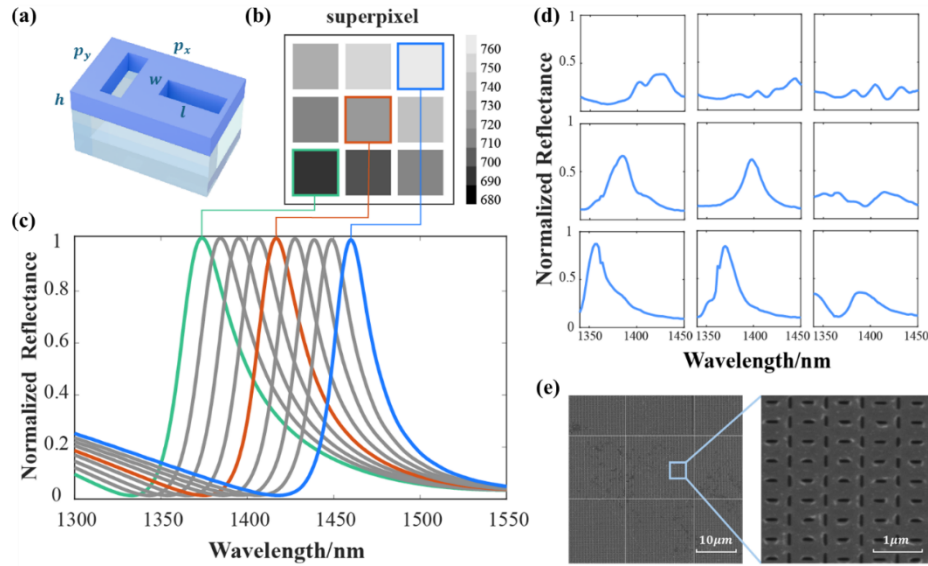


Fig. 3. Structures and spectral reflectance. (a) Render image of the meta-atom composed of two orthogonal rectangular airholes, where $h = 250\text{ nm}$, $w = 100\text{ nm}$, $l = 300\text{ nm}$, $p_y = 400\text{ nm}$, p_x is a tuned parameter scaling from 680 nm to 760 nm with a step of 10 nm . (b) Schematic of the super pixel (microspectrometer). A larger grayscale value indicates a denser distribution and smaller period. (c) Simulation reflectance spectra of nine types of meta-atom arrays. Resonance positions of the colored curves correspond to the respective reflectance images in (b). (d) Experimentally measured reflectance spectra extracted from the calibration. (e) Scanning electron microscopy (SEM) images for a part of superpixel (light) and meta-atoms (right).

We calculated the reflection spectrum of nine types of the meta-atom arrays constituting the metasurface by using finite-difference-time-domain (FDTD) method [Fig. 3(c)]. Due to the BICs of different meta-atom arrays, the range of the reflection peak covers $1350\text{--}1450\text{ nm}$ as the period changes. Figure 3(d) shows spectral reflectance curves of experimental calibration, whose subplot corresponding one-to-one with the filter pixels in Fig. 3(b). In Fig. 3(c), the grey curves show the simulated spectra, whereas the green, orange and blue spectra correspond to the filters highlighted in Fig. 3(b). The drift of resonance peak position and the inconsistency of curve shapes are attributed to error in geometric precision dimensions caused by fabrication limitations. That is why we chose the broadband method based on convex optimization, rather than the narrowband method based on filter arrays. Besides, our non-coincident resonant frequency of BIC metasurface arrays can lead to low cross-correlations between their reflection spectra. This method resulted in a higher spectral resolution than the initial narrowband filtering method.

2.3. Experiment details

We used the optical setup shown in Fig. 4(a) for calibration, spectral curve measurements, and spectral imaging measurements [38]. The experiment is based on a microscope system with an objective magnification of 20x. The optical path includes two beams of light, one red and one blue. When we perform calibration, only the red path emitted from laser1 (NKT Photonics SuperK SELECT) needs to be used. A light wave emitted from super-continuum Laser1 is split by a beam splitter (BS). The reflection part is collected by a power meter or a commercial spectrometer as reference. The transmission part to obtain the results of a single measurement passes through a linear polarizer (LP), a 4-f system (f_1 of lens1 = 200 mm, f_2 of lens2 = 75 mm), and then is split by the second BS. Beam modulated by metasurface passes through an iris twice to ensure normally incidence, and is received by an infrared camera at last. In the calibration, the wavelength bands were set from $\lambda = 1340$ nm to 1450 nm with a sampling interval $\Delta\lambda$ of 2.5 nm (a total of 44 bands). Figure 4(b) shows experimentally measured three examples of calibration. We can see that when light of different wavelengths is incident, only one filter pixel on the metasurface highlights while other area is dark, which is in good agreement with our theory. Laser2 (NKT Photonics SuperK COMPACT) is a broadband white light source that outputs 450-2400 nm continuous laser, used in combination with a filter. While in measuring spectral curves, turn on Laser1 and Laser2 simultaneously to generate more complex waveforms or overlapping images carrying different spectral information as test data. It should be noted that after being reflected by Plane mirror the blue path from Laser2 must be fully aligned with the red one in order to measure accurate reference spectrum and reconstructed spectrum.

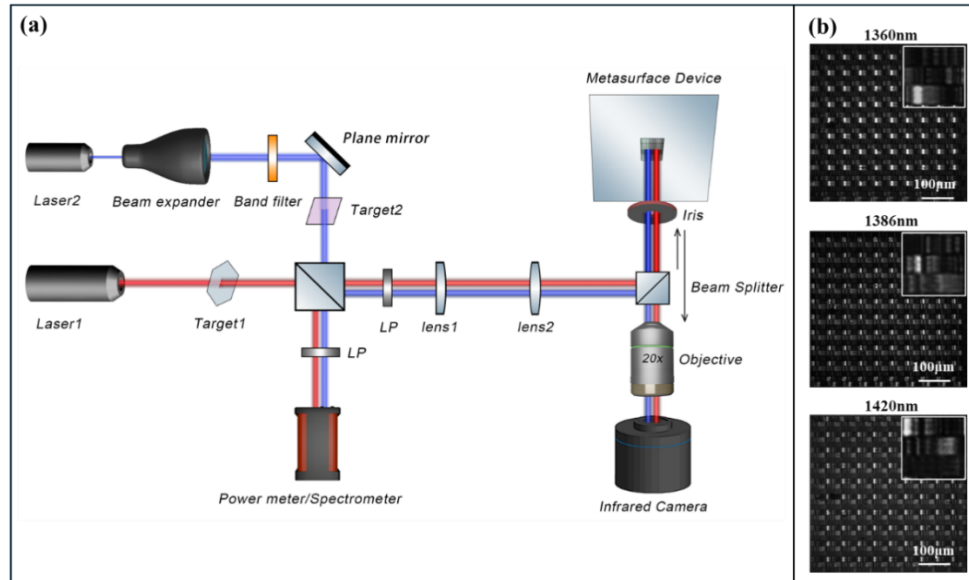


Fig. 4. (a) Experiment setup for BIC metasurface spectral imaging (b) the real captured images of metasurface devices when illuminated with light of wavelengths 1360 nm, 1386 nm and 1420 nm. The top right corner shows a magnified view of a superpixel comprises $3 \times 3 = 9$ metasurface units.

3. Result

3.1. Algorithm description

The mathematical model for our system has been presented in Eqs. (1) and (2). Encoded light from the targets is captured as a single monochromatic image b by an infrared camera (QHYCCD 990). To recover the multispectral datacube from the grayscale 2D measurement, we must solve an underdetermined inverse problem. Since the tested scene is usually spectrally sparse, we used l_1 regularization. The reconstruction process under sparsity assumption can be written as:

$$L = \arg \min_f \|b - Af\|_2^2 + \tau \|TV(f)\|_1 \quad (4)$$

where A is the calibrated matrix of 9×44 based on the designed metasurface patterns, f is the reconstructed spectra and b is detected values matched with spectral filter array and is reshaped to a column vector. Owing to the magnification of the microscope system, one filter pixel is mapped by dozens of sensor pixels, so that we take the center pixel of each filter region as a valid result of measurement. We consider contained optimization problem with convex and constraints to solve the inverse problem. Specifically, solving Eq. (4) minimizes the loss function L to find the reconstructed spectra by CVXPY based on ECOS solver. CVXPY is an open source Python-embedded modeling language for convex optimization problems [39]. We use a weighted total variation constraint on the spectrum, τ is the regularization coefficient compared with the first fidelity term and TV represents total variation [40,41]. TV loss is a commonly used regularization term in image processing, which penalizes the gradient of the image to remove noise and preserve edges and other information within the image. In optimization, adjusting the weight of the TV loss can impact the quality and stability of the reconstructed result. When the weight τ of TV is small, it will maintain more sharp edges, but the reconstruction results are also more susceptible to sensor noise, leading to an unstable reconstruction. Conversely, a larger τ makes the reconstructed image smoother, allowing for more accurate spectral profile, which is suitable for reconstructing broadband spectra. Therefore, by appropriately adjusting the weight of the TV loss, it is possible to effectively remove noise while preserving image details, resulting in a clear and stable reconstructed image.

In the laboratory, researchers often pay more attention to the difference in the shape of the spectrum than to the difference in the amplitude of the spectrum. Therefore, the index of fidelity is introduced to quantitatively analyze the reconstruction effect, which is defined as:

$$F(X, Y) = \frac{Y^T X}{\|X\| \|Y\|} \quad (5)$$

where X and Y are the normalized spectral intensity of reference spectra and reconstructed spectra, respectively, $\|\cdot\|$ means l_2 -Normalized. This kind measurement method of spectral similarity based on projection, essentially for the cosine similarity. Since the amplitude and shape of the spectrum correspond to the length and direction of the vector in Euclidean space, the multiplicative factor only causes changes in the length of the vector, but does not change the direction of the vector. Thus, this method is sensitive to spectral shape differences, but not to spectral amplitude differences. This helps researchers to identify some special waveforms and realize analyte detection in scientific research.

3.2. Spectral curve reconstruction

We used the device in Fig. 4(a), with laser 1 emitting a narrowband spectrum and laser 2 emitting a broadband spectrum, to test the performance of the proposed BIC metasurface spectrometer consisting of nine filter pixels. Figure 5(a) denotes narrowband wavelength measurement of monochromatic light sources over a wavelength range of 1340-1450 nm with a step of

10 nm, where the blue lines represent the reference spectra directly measured by commercial spectrometers and the black lines represent our reconstructed spectra. We calculated the average fidelity of 12 narrow band spectra as 98.05%. We then measured other cases using a combination of monochromatic light source and white light source with bandpass filters, such as broadband and narrowband superimposed incidence [Fig. 5(b)] and sets of narrowband spectra with two, three, four peaks [Fig. 5(c-e)]. The corresponding spectral fidelity is 97.25%, 95.98%, 95.91% and 92.83%, respectively.

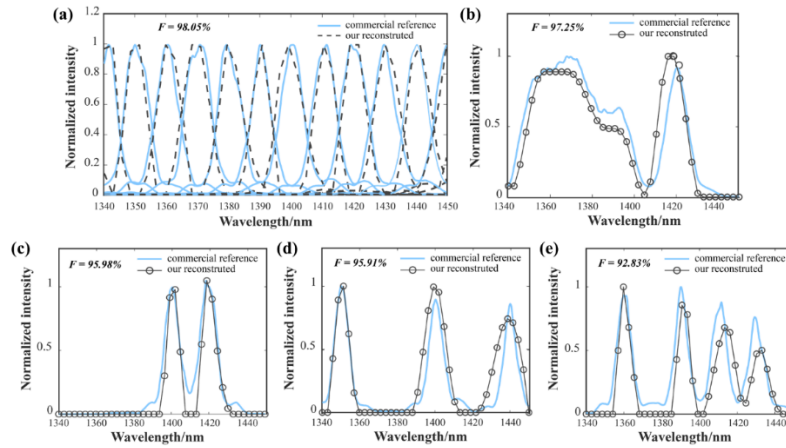


Fig. 5. Typical spectra curve reconstruction of the proposed device. (a)Reconstructed results of 12 narrowband spectra from of monochromatic light sources, where the blue lines represent the reference spectra directly measured by commercial spectrometers and the black lines represent our reconstructed spectra. (b)Reconstruction of a broadband and narrowband combined spectrum. (c), (d) and (e)Reconstructed results of three combinations of narrowband spectra: double-peak, triple-peak and quadruple-peak spectrum.

3.3. Spectral imaging

Besides, we proved the potential of our BIC metasurface spectrometer for multispectral imaging. In the previous experiments, the size of each microspectrometer was $50\mu\text{m}$ on a side, which consisted of 9 meta-atom arrays working as filter masks. Here, we repeatedly distributed these identical microspectrometers into a 10×10 array for a multispectral imager with 100 pixels. In this part, our target was formed by projecting the figure of equal sign (indicated by red light) and square (indicated by blue light) on the metasurface, as shown in Fig. 6(a). The equal sign was illuminated by light with a wavelength of 1400 nm and the square was illuminated by light with a wavelength of 1360 nm. The mixed light(indicated by purple light) is modulated by the BIC metasurface and all spatial and spectral information was recorded in a single image [Fig. 6(b)]. These two shapes overlap spatially and are hardly distinguished using a conventional monochrome infrared camera. To verify the performance in spectra separation, we use rectangular boxes of different colors to select different regions for spectral reconstruction. As Fig. 6(c) shows, the equal sign area is selected in blue, the square area is selected in cyan, and the overlapping area is selected in orange. Three reconstructed spectrum of selected area are marked by the same color as boxes and double peaks only appears in overlapping area. We also reconstructed all image pixels (i.e. superpixels) by means of traversal to obtain spatial information for each wavelength. Figure 6(d) manifests ten images from 1350 nm to 1440 nm, at increments of 10 nm. The results show that in the 1360 nm and 1400 nm channels, there can be almost no crosstalk and the two

shapes can be readily distinguished. This proves the ability of our proposed device to spectral imaging.

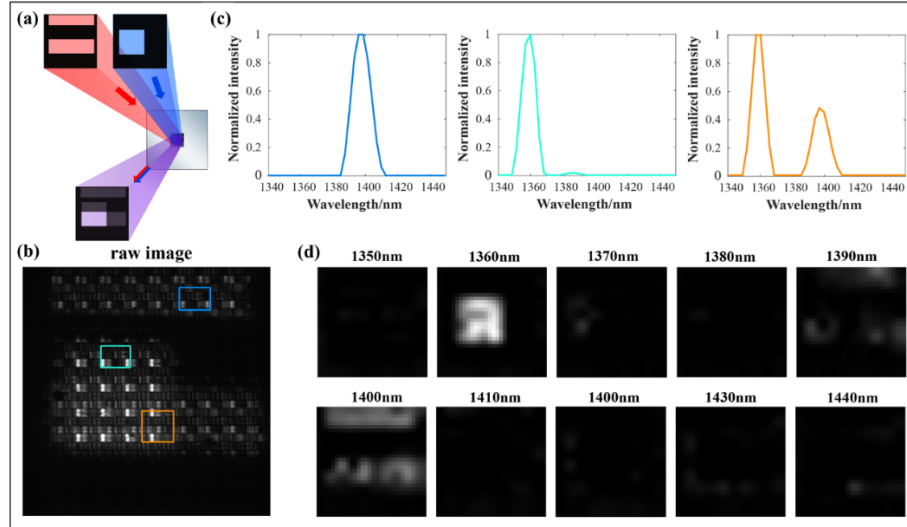


Fig. 6. Spectral imaging experiment. (a) Brief schematic diagram of the imaging target. The target is formed by projecting the figure of equal sign(indicated by red light) and square(indicated by blue light) on the metasurface. The equal sign area is selected in blue, the square area is selected in cyan, and the overlapping area is selected in orange. (b) Monochromatic image captured by an infrared camera before reconstruction. (c) Reconstruction results of the selected area in (b). (d) Ten images from the reconstructed data cube with 40 wavelengths.

4. Discussion

Table 1 is comparison of this work with a series of demonstrated micro-spectrometer platforms, both based on broadband filters array and computational reconstruction. Principles refer to the specific materials or optics utilized by these spectrometers in order to achieve filtering. While these compared work in the visible light band, our quasi-BIC metasurface works in the near infrared band, and the band can be customized through the design of materials and periods. Micro-spectrometer footprint is the plane size of filters array, which is regarded as an imaging pixel. This value may not equal to the spatial resolution due to the use of some algorithmic strategies, such as in Ref. [28] and [45]. The definition of spectral resolution is different in these articles. The “measured” marked in the table represents the minimum separation between two indistinguishable peaks in the target spectrum. The unmarked means that this test has not been done, and the value is spectral reconstruction accuracy calculated based on the reconstruction bandwidth and bands number. Fidelity is not the only index for evaluating the quality of reconstruction of all articles.

The advantages of quasi-BIC metasurfaces are reflected in their potential for multi-functional applications. Compared with other materials, quasi-BIC is easy to design, flexible and adjustable, and only changing a certain parameter can get a regular response. Not only is there a variable response band, but the bandwidth can also be regulated by its Q factor. We have chosen polarization-dependent dimer units, whose functions are not limited to the spectral imaging mentioned in this paper. By controlling the steering angle of the structure, polarization selection

Table 1. Comparison between our work and other reported works

References	Principles	Operational Wavelength	Micro-spectrometer Footprint	Spectral resolution	Reconstruction quality
Ref. [42]	Photonic crystal slabs	450-750nm	$80 \times 80 \mu\text{m}^2$	1nm	/
Ref. [28]	Reconfigurable metasurfaces	550-750nm	$210 \times 210 \mu\text{m}^2$	0.8 nm(measured)	>98%
Ref. [43]	Quantum dot	390-690nm	$8.5 \times 6.8 \text{mm}^2$	3.2 nm(measured)	/
Ref. [44]	Single nanowire	490-630nm	$0.5 \times 75 \mu\text{m}^2$	10 nm(measured)	/
Ref. [45]	Fabry-pérot filters	450-650nm	$38.5 \times 38.5 \mu\text{m}^2$	10nm	/
This work	q-BIC metasurfaces	1340-1450nm	$50 \times 50 \mu\text{m}^2$	2.5nm	>92%

and geometric phase regulation can be realized, so as to realize metalens and polarization-spectrum detection, which will be reflected in future work.

5. Conclusion

In conclusion, we present a metasurface device based on the quasi-BIC principle that can be used as an infrared spectrometer and imager. Our device is fabricated by the deposition of a α -Si thin film on a SiO_2 substrate and electron beam exposure and subsequent steps. The performance of the device is simulated by FDTD and verified experimentally. We reconstructed the spectrum between 1340 nm and 1450 nm with a spectral resolution of 2.5 nm using a CVX optimization algorithm under sparse assumptions. For the reconstruction of monochromatic narrowband spectral curves, the average fidelity is 98.05%. The reconstruction fidelity of other complex waveforms is all over 92%. In addition, we show that it is possible to distinguish images with spatial overlap in different wavelength bands. It overcomes the shortcomings of long-time scanning, high cost and large volume of traditional methods, and demonstrates the broad prospect of low-cost, rapid-imaging portable spectrometers in agriculture, product safety and biosensing.

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